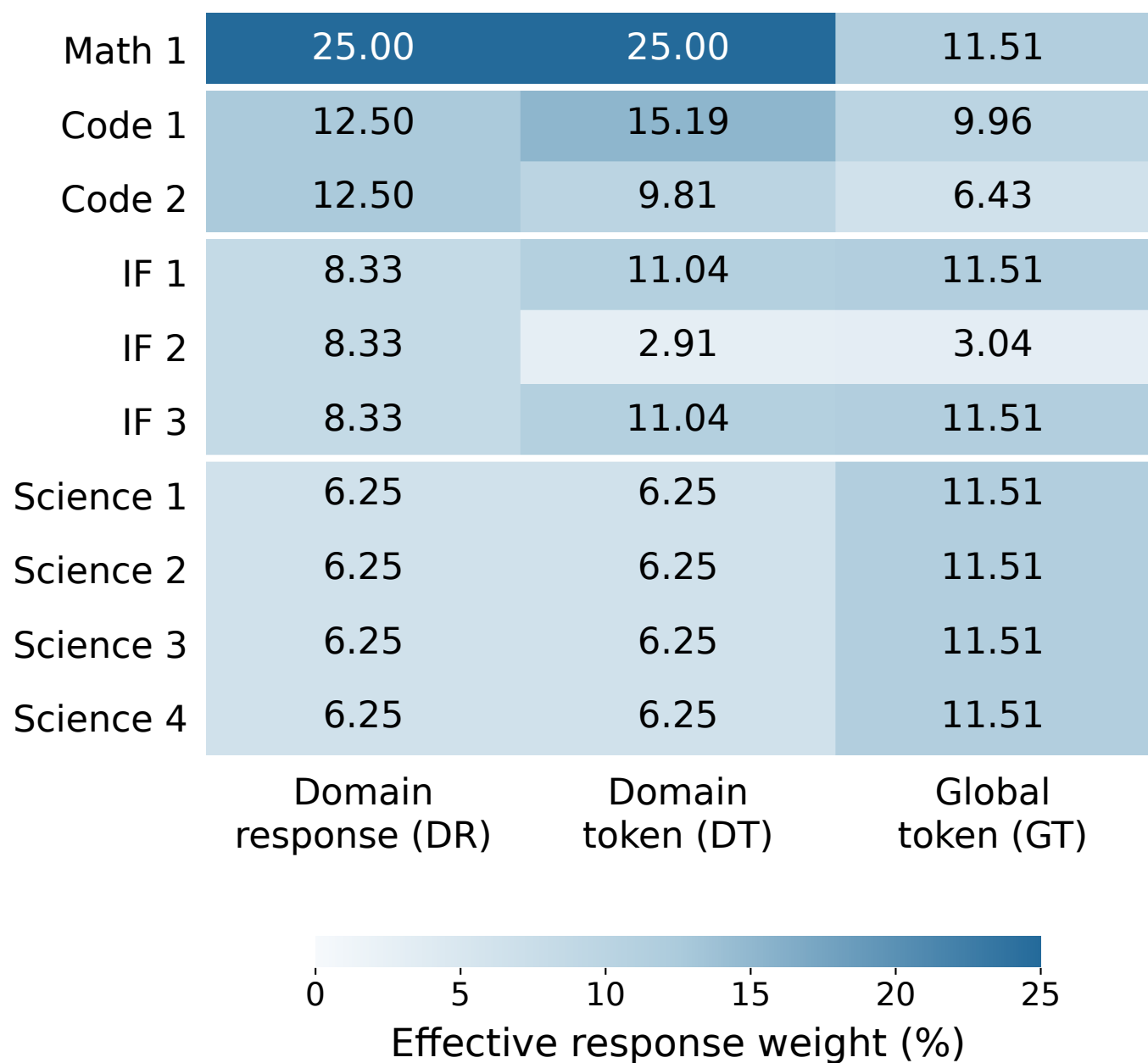
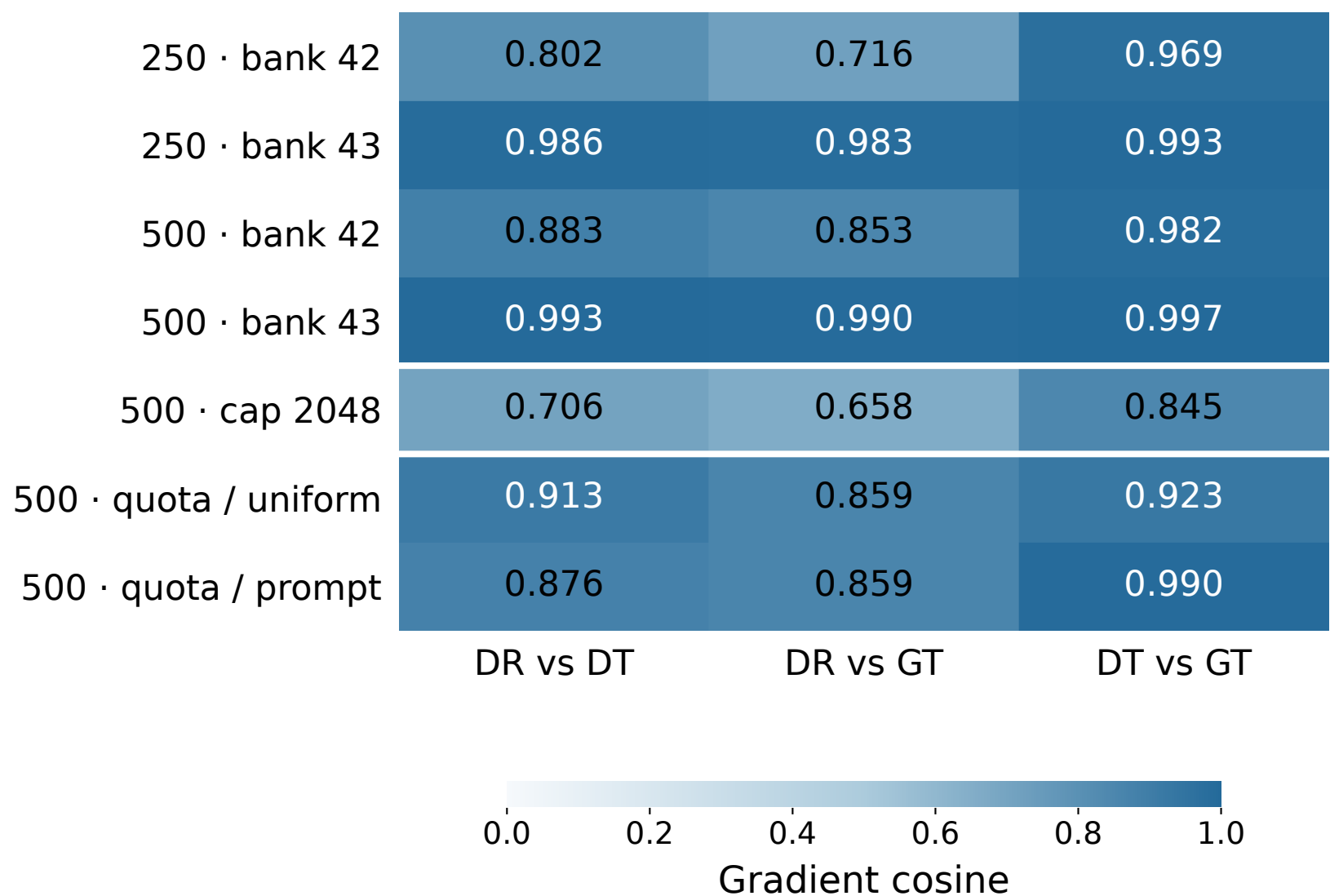
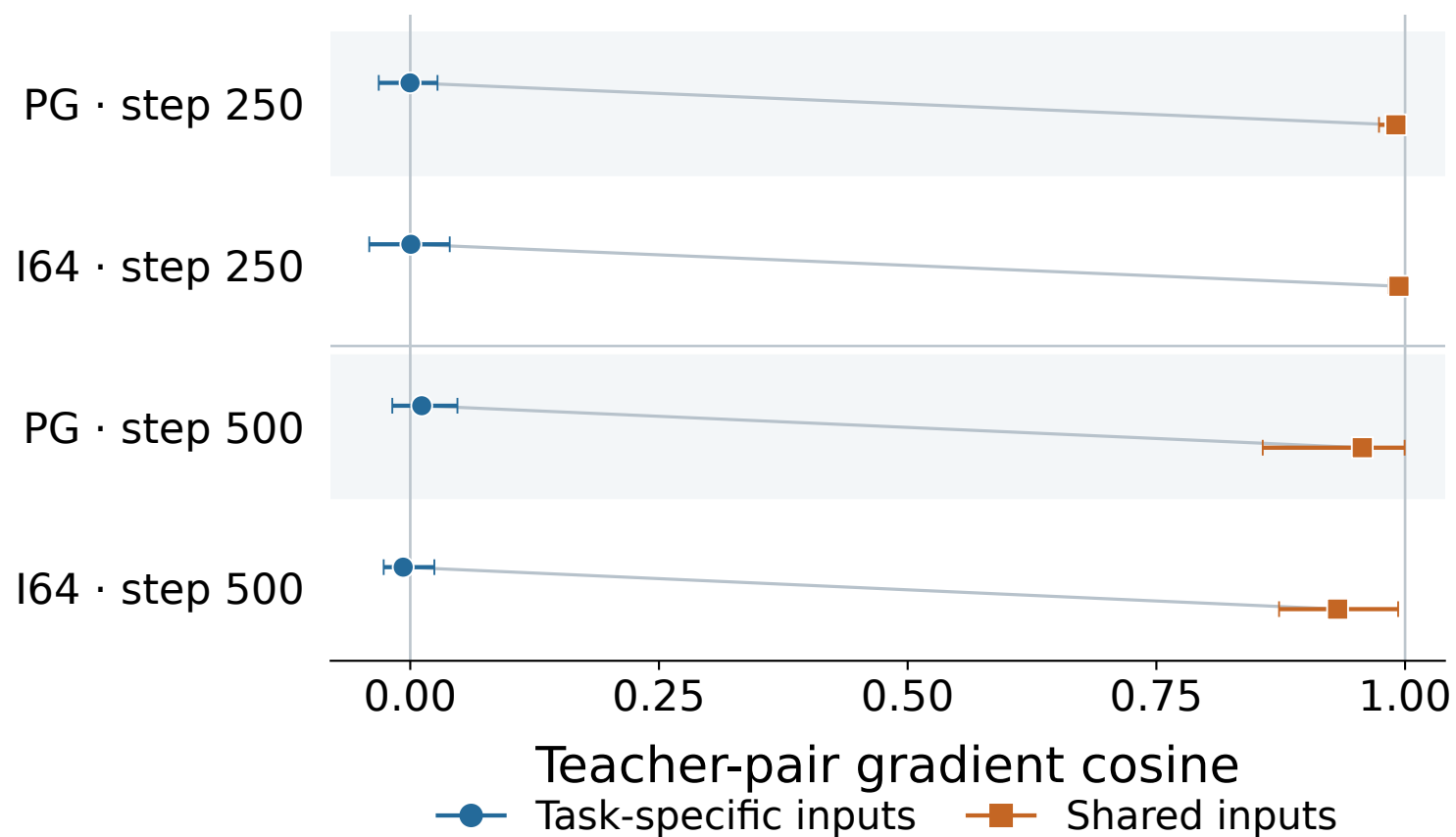
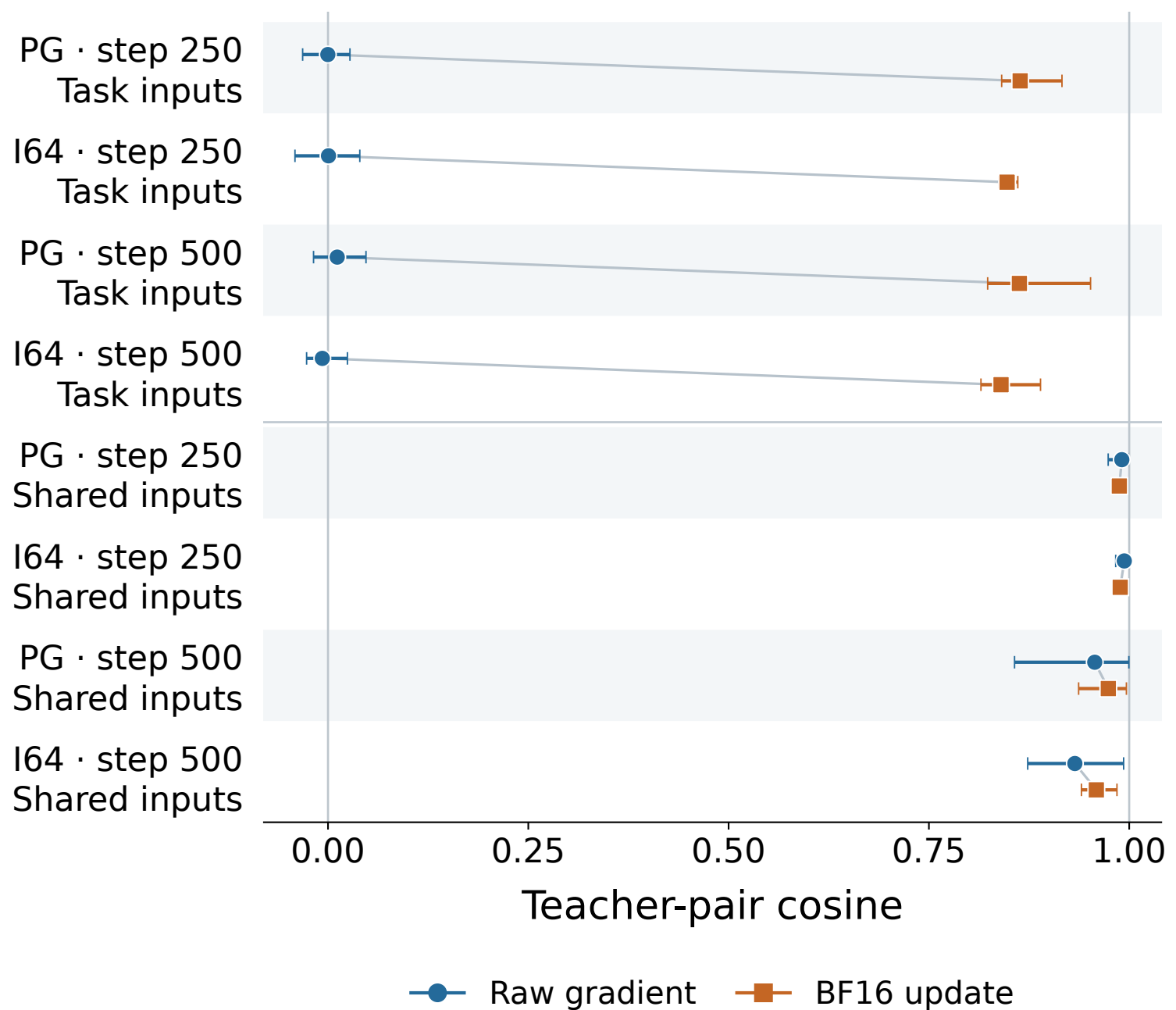
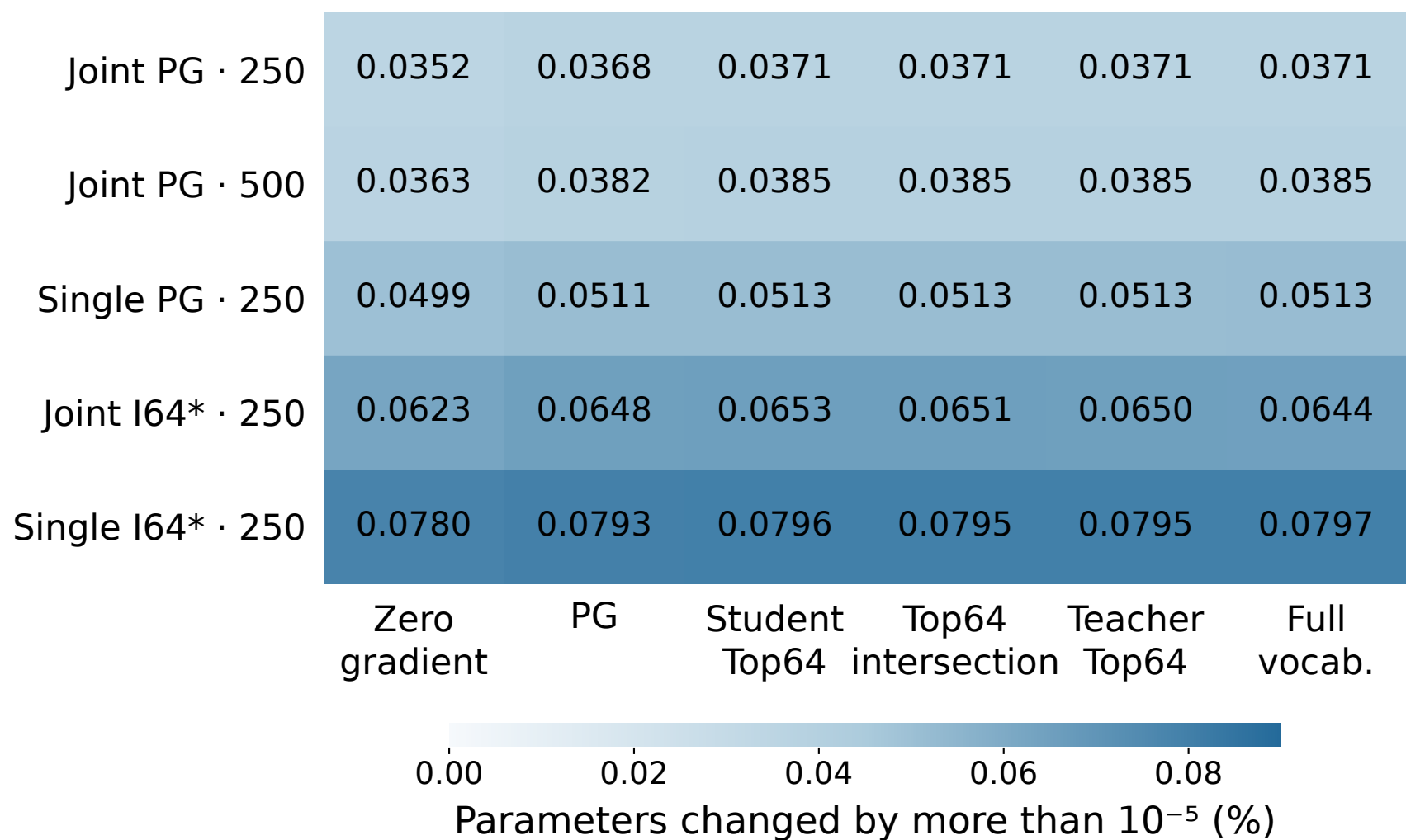






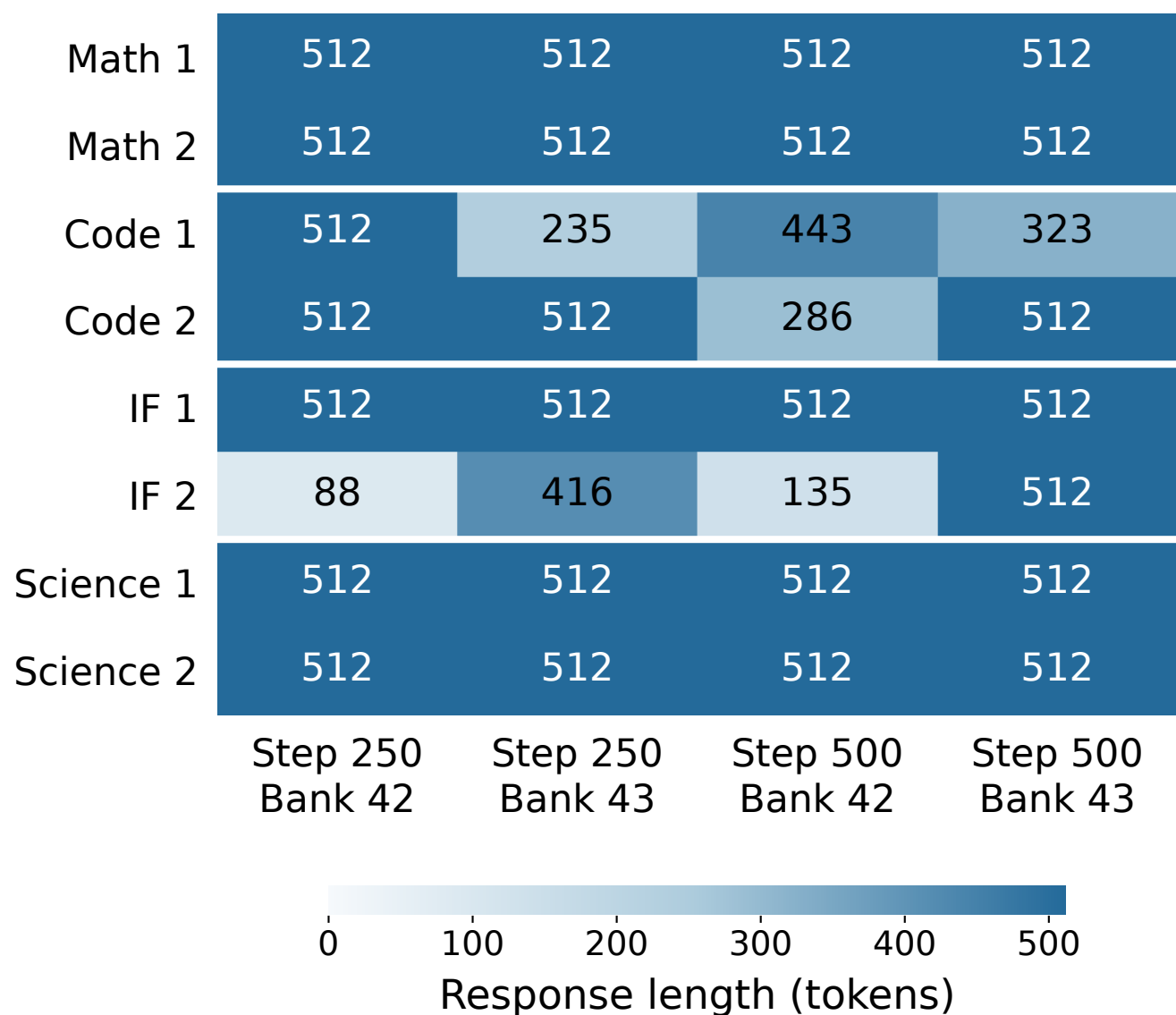


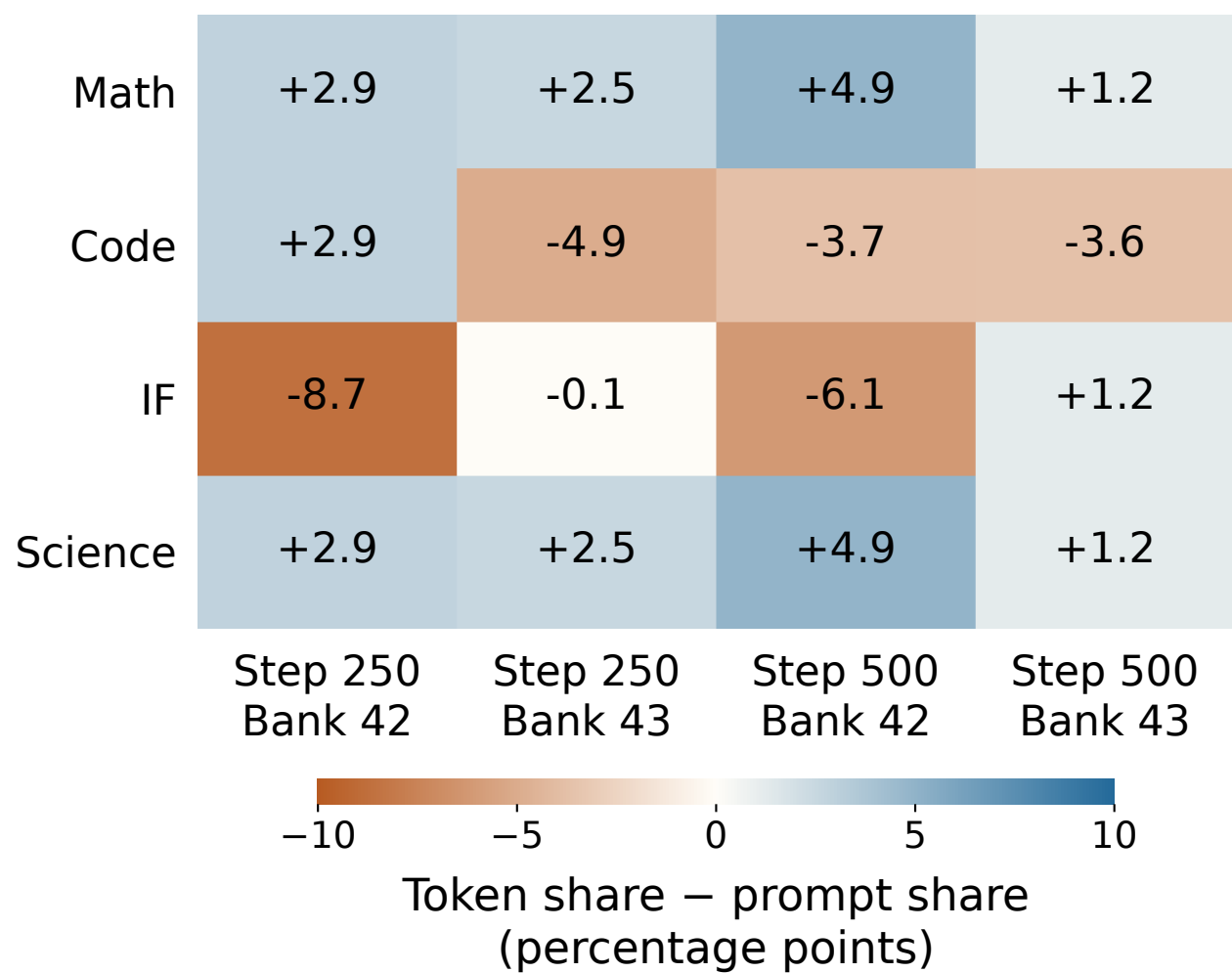




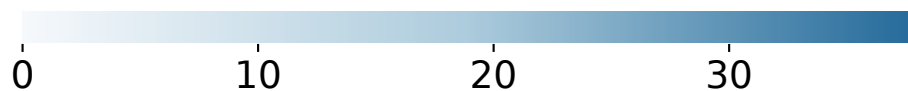
Joint PG · 250	Gradient	0.000	0.659	0.650	0.642	0.675
	Update	0.986	1.005	1.004	1.004	1.005
Joint PG · 500	Gradient	0.000	0.820	0.757	0.746	0.812
	Update	0.981	1.003	1.003	1.003	1.004
Single PG · 250	Gradient	0.000	0.171	0.173	0.173	0.171
	Update	0.992	1.002	1.002	1.002	1.003
Joint I64* · 250	Gradient	0.000	0.424	0.381	0.347	0.452
	Update	0.984	1.004	1.002	1.002	0.997
Single I64* · 250	Gradient	0.000	0.136	0.090	0.074	0.416
	Update	0.993	1.002	1.001	1.001	1.002
		Zero gradient	Student Top64	Top64 intersection	Teacher Top64	Full vocab.







250 · bank 42	0.00	0.00	38.04	0.00
250 · bank 43	0.00	5.23	1.42	0.00
500 · bank 42	0.00	4.53	19.15	0.00
500 · bank 43	0.00	2.82	0.00	0.00
500 · cap 2048	0.62	4.53	27.13	0.83
500 · quota / uniform	0.00	4.53	13.59	0.00
500 · quota / prompt	0.00	4.53	13.59	0.00
	Math	Code	IF	Science



Length-weighting correction (L2)

250 · bank 42	+0.59	-0.38	+0.74
250 · bank 43	-2.90	-1.24	+0.56
500 · bank 42	-0.81	-0.65	-1.21
500 · bank 43	-1.30	-2.36	-2.67
500 · cap 2048	+0.98	+0.02	-0.08
500 · quota / uniform	+0.29	+0.12	+0.01
500 · quota / prompt	+0.72	-1.04	-1.35
	DR	DT	GT



Extra KL decrease vs zero-gradient
(millinats)

